

DriveFace: A Cross-Spectral Through-Glass Face Dataset for On-the-Move Vehicular Border Control

Anjith George¹, Luis S. Luevano¹, Alain Komaty¹, Zeina Al Amine¹, Vidit Vidit¹, Sébastien Marcel¹²

¹Idiap Research Institute ²University of Lausanne (UNIL)

Motivation

On-the-move border control must identify travellers *inside their vehicles*. Existing benchmarks ignore **motion blur**, **variable illumination**, **occlusion**, **window tint** and **cross-spectral (RGB→NIR)** matching **through automotive glass**.

Contributions

- **DriveFace dataset**: 70 subjects, 2 sessions; visible smartphone pre-enrollment paired with in-vehicle NIR probes through glass.
- **Cross-spectral, through-glass** RGB–NIR matching at varying tint, pose, and speed — stationary and moving.
- **DriveFace-PAD**: print, replay, and mask attacks across tint and illumination.
- **Standardized FR + PAD protocols** with rich metadata (tint, illumination, pose, speed) and strong baselines.

70

subjects

2

sessions

5

tint levels

18–85

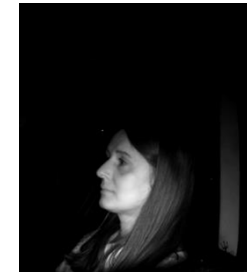
age



Setup



Outdoor 45°



Indoor car



Sim. tint

Demographics

Demographically diverse and gender-balanced (**45.7% female / 54.3% male**), spanning ages 18–85. Apparent skin tone grouped light / medium / dark = **51 / 11 / 6** subjects, supporting fairness-aware evaluation.

Results & Key Findings

Face recognition. **xEdgeFace** adapts EdgeFace cross-spectrally; the tinted-glass *simulation* is the hardest setting.

	Model	AUC	EER	VR@1%	Rank-1
Outdoor	AdaFace	99.45	2.69	96.01	97.42
	LVFace	98.50	5.26	90.17	92.26
	EdgeFace	98.38	5.37	91.35	93.29
	xEdgeFace	98.94	4.18	93.12	94.22
Simul.	AdaFace	96.23	8.26	88.27	90.46
	LVFace	95.19	11.42	79.68	83.95
	EdgeFace	94.09	11.73	82.93	84.72
	xEdgeFace	96.03	8.09	88.63	89.68
Ind. Car	AdaFace	98.59	3.11	96.24	97.11
	LVFace	98.52	5.29	90.23	93.27
	EdgeFace	98.75	4.48	93.92	93.34
	xEdgeFace	97.98	3.47	95.95	96.16

VR@FAR=1% best per protocol in **blue**.

Tint effect (AUC): T05 98.4% → **clear 100.0%** — accuracy scales with visible-light transmission.

Presentation attack detection. **DriveFace-PAD** — known (*grandtest*) attacks are detected reliably; *unseen* attacks reveal a generalization gap.



Laser-matte



Silicone



Paper mask



Resin mask

Protocol	Best model	ACER
grandtest	EfficientNet-B0	0.5
unseen_mask	DinoV2 (fc)	26.2
unseen_print	DeepPixBiS	18.4

Best ACER (%) per protocol; unseen attacks up to 45.8%.

Key Findings

- Robust FR: **96–97% Rank-1** outdoors & indoor-car despite the RGB–NIR gap; tinted-glass simulation is hardest.
- Cross-spectral adaptation (**xEdgeFace**) improves over EdgeFace.
- PAD strong on known attacks (**0.5% ACER**), hard on unseen ones (**up to 45.8%**).

Paper, code & dataset: www.idiap.ch/paper/driveface

